Theorem 1 (Equivalence of exact privacy curves for additive symmetric noise)**.** *Let*

$$M(x) = A(x) + Z,$$

where $A : \mathcal{X} \to \mathbb{R}^d$ is deterministic and $Z \sim \nu$, with ν a centered symmetric probability distribution on $\mathbb{R}^d$, i.e.

$$\nu(B) = \nu(-B) \qquad \text{for every Borel set } B \subseteq \mathbb{R}^d.$$

Fix neighboring datasets x, x', and define

$$a := A(x), \qquad a' := A(x'),$$

$$P := \mathcal{L}(a + Z), \qquad Q := \mathcal{L}(a' + Z).$$

Assume $P \ll Q$ (equivalently $Q \ll P$ in this setting). Let

$$L := \frac{dP}{dQ}.$$

Then the following data determine one another:

1. *the full exact Rényi-DP curve $\{D_\alpha(P\|Q) : \alpha > 1\}$;*
2. *the full exact f-DP curve;*
3. *the exact (ε, δ)-DP curve for $\varepsilon \geq 0$.*

Equivalently, in this symmetric additive-noise setting, the one-sided exact privacy profile $\{\delta(\varepsilon) : \varepsilon \geq 0\}$ already determines the full two-sided profile $\{\delta(\varepsilon) : \varepsilon \in \mathbb{R}\}$, and hence determines the Rényi-DP and f-DP curves. The converse implications also hold.

Proof. We proceed in several steps.

Step 1: Reflection symmetry exchanges P and Q.

Define the midpoint

$$m := \frac{a + a'}{2},$$

and the reflection

$$T(y) := 2m - y = a + a' - y, \qquad y \in \mathbb{R}^d.$$

Then T is a measurable involution: $T \circ T = \mathrm{id}$.

We claim that

$$T_\# P = Q, \qquad T_\# Q = P.$$

Indeed, if $Y = a + Z \sim P$, then

$$T(Y) = a + a' - (a + Z) = a' - Z.$$

Since $Z \overset{d}{=} -Z$ by symmetry of ν, we have

$$a' - Z \overset{d}{=} a' + Z \sim Q.$$

Hence $T_\# P = Q$. By symmetry the reverse identity $T_\# Q = P$ also holds.

Step 2: The privacy profile is symmetric under swapping P and Q.

Recall the exact privacy profile

$$\delta_{P,Q}(\varepsilon) := \sup_E \big(P(E) - e^{\varepsilon} Q(E)\big), \qquad \varepsilon \in \mathbb{R}.$$

For any measurable E,

$$P(E) - e^{\varepsilon} Q(E) = Q(T^{-1}E) - e^{\varepsilon} P(T^{-1}E).$$

Taking the supremum over E, and using that T^{-1} is a bijection on measurable sets, yields

$$\delta_{P,Q}(\varepsilon) = \delta_{Q,P}(\varepsilon) \qquad \forall \varepsilon \in \mathbb{R}.$$

Thus the privacy profile is invariant under exchanging P and Q.

Step 3: Negative ε is determined by positive ε.

For arbitrary probability measures P, Q, one has the identity

$$\delta_{P,Q}(-\varepsilon) = 1 - e^{-\varepsilon} + e^{-\varepsilon} \delta_{Q,P}(\varepsilon), \qquad \varepsilon \in \mathbb{R}.$$

Indeed,

$$\delta_{P,Q}(-\varepsilon) = \sup_E \big(P(E) - e^{-\varepsilon} Q(E)\big).$$

Writing $F = E^c$, we get

$$\begin{aligned} P(E) - e^{-\varepsilon} Q(E) &= 1 - P(F) - e^{-\varepsilon}(1 - Q(F)) \\ &= 1 - e^{-\varepsilon} + \big(e^{-\varepsilon} Q(F) - P(F)\big) \\ &= 1 - e^{-\varepsilon} + e^{-\varepsilon}\big(Q(F) - e^{\varepsilon} P(F)\big). \end{aligned}$$

Taking the supremum over E, equivalently over F, proves the identity.

Now apply Step 2:

$$\delta_{Q,P}(\varepsilon) = \delta_{P,Q}(\varepsilon).$$

Therefore

$$\boxed{\delta(-\varepsilon) = 1 - e^{-\varepsilon} + e^{-\varepsilon} \delta(\varepsilon), \qquad \varepsilon \geq 0.}$$

Hence knowledge of $\delta(\varepsilon)$ for all $\varepsilon \geq 0$ uniquely determines $\delta(\varepsilon)$ for all $\varepsilon < 0$. So the one-sided privacy profile determines the full two-sided profile.

Step 4: The full two-sided privacy profile determines the law of the likelihood ratio.

Let

$$L := \frac{dP}{dQ}.$$

As usual,

$$\delta(\varepsilon) = \mathbb{E}_Q[(L - e^{\varepsilon})_+], \qquad \varepsilon \in \mathbb{R}.$$

Equivalently, if

$$g(u) := \mathbb{E}_Q[(L - u)_+], \qquad u > 0,$$

then

$$\delta(\varepsilon) = g(e^{\varepsilon}).$$

For every $u > 0$, g is convex and has one-sided derivatives

$$g'_-(u) = -Q(L \geq u), \qquad g'_+(u) = -Q(L > u).$$

Thus knowing $g(u)$ for all $u > 0$ determines the tail probabilities of L, hence the law of L under Q.

Since Step 3 shows that the one-sided profile $\{\delta(\varepsilon) : \varepsilon \geq 0\}$ determines the full profile $\{\delta(\varepsilon) : \varepsilon \in \mathbb{R}\}$, it follows that in the present symmetric setting, the one-sided profile determines the law of L.

Step 5: The law of L determines the full exact Rényi-DP curve and the full exact f-DP curve.

For every $\alpha > 1$,

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \mathbb{E}_Q[L^\alpha].$$

Hence the law of L determines the full Rényi-DP curve.

Likewise, by the Neyman–Pearson lemma, the exact f-DP / tradeoff curve is determined by threshold tests in L, hence is determined by the law of L.

Therefore the one-sided exact privacy profile determines both the full exact Rényi-DP curve and the full exact f-DP curve.

Step 6: Converse implications.

The full exact Rényi-DP curve determines the law of L: indeed, for $\alpha > 1$,

$$\mathbb{E}_Q[L^\alpha] = e^{(\alpha-1)D_\alpha(P\|Q)},$$

so the moment generating function of $\log L$ is known on $(1, \infty)$, which determines its distribution uniquely.

The full exact f-DP curve also determines the law of L, by Neyman–Pearson.

Once the law of L is known, the exact privacy profile is given by

$$\delta(\varepsilon) = \mathbb{E}_Q[(L - e^\varepsilon)_+].$$

In particular, this determines the one-sided profile $\varepsilon \geq 0$.

Thus all three objects determine one another. □